Year 11 Foundation

REVISION PACK

Graphs 5

Name: _____

Class: _____

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding the equations

Solve the simultaneous equations

$x + y = 9$

$x - y = 3$

Step 1	$+y$ and $-y$: add	$2x = 12$
Step 2	Solve	$x = 6$
Step 3	Put x in	$6 + y = 9$
Step 4	Solve	$y = 3$

Practice: adding the equations

1.1 Solve the simultaneous equations

$x + y = 12$

$x - y = 4$

1.2 Solve the simultaneous equations

$x + y = 11$

$x - y = 7$

1.3 Solve the simultaneous equations

$x + y = 12$

$x - y = 2$

1.4 Solve the simultaneous equations

$x + y = 13$

$x - y = 7$

1.5 Solve the simultaneous equations

$x + y = 7$

$x - y = 5$

1.6 Solve the simultaneous equations

$x + y = 15$

$x - y = 7$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: subtracting the equations

Solve the simultaneous equations

$4x + y = 17$

$x + y = 8$

Step 1	$+y$ and $+y$: subtract	$3x = 9$
Step 2	Solve	$x = 3$
Step 3	Put x in	$3 + y = 8$
Step 4	Solve	$y = 5$

Practice: subtracting the equations

1.7 Solve the simultaneous equations

$2x + y = 13$

$x + y = 8$

1.8 Solve the simultaneous equations

$4x + y = 14$

$x + y = 5$

1.9 Solve the simultaneous equations

$3x + y = 19$

$x + y = 7$

1.10 Solve the simultaneous equations

$5x + y = 17$

$x + y = 9$

1.11 Solve the simultaneous equations

$2x + y = 20$

$x + y = 12$

1.12 Solve the simultaneous equations

$3x + y = 21$

$x + y = 13$

Graphs 5 · Simultaneous equations (Lesson 3.1)

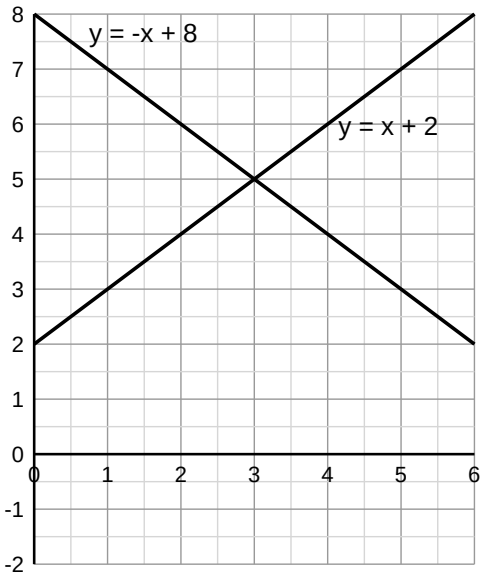
Worked example: solving using graphs

The graphs of $y = x + 2$ and $y = 6 - x$ are drawn on a grid. They cross at (2, 4). Use the graphs to solve the simultaneous equations.

Step 1	Where the lines cross	$(2, 4)$
Step 2	Answer	$x = 2,\ y = 4$

Practice: solving using graphs

1.13



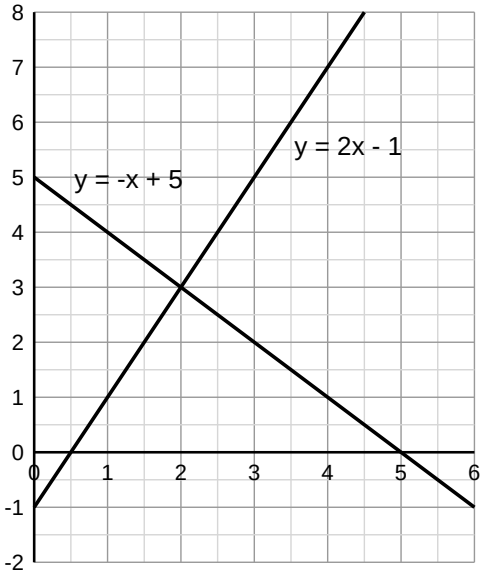
The graphs of $y = x + 2$ and $y = -x + 8$ are drawn on the grid.

Use the graphs to solve the simultaneous equations

$y = x + 2$

$y = -x + 8$

1.14



The graphs of $y = 2x - 1$ and $y = -x + 5$ are drawn on the grid.

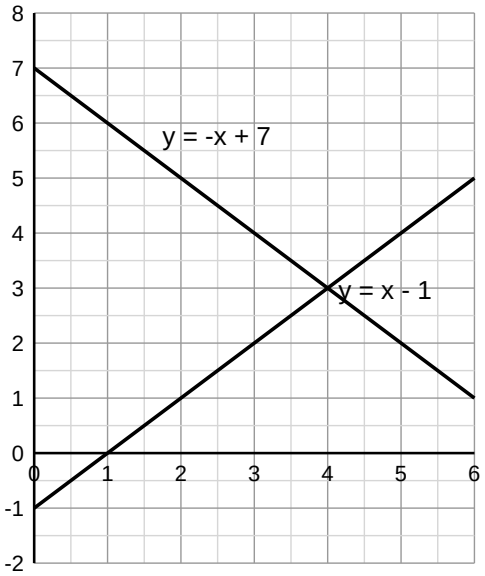
Use the graphs to solve the simultaneous equations

$y = 2x - 1$

$y = -x + 5$

Practice: solving using graphs (continued)

1.15



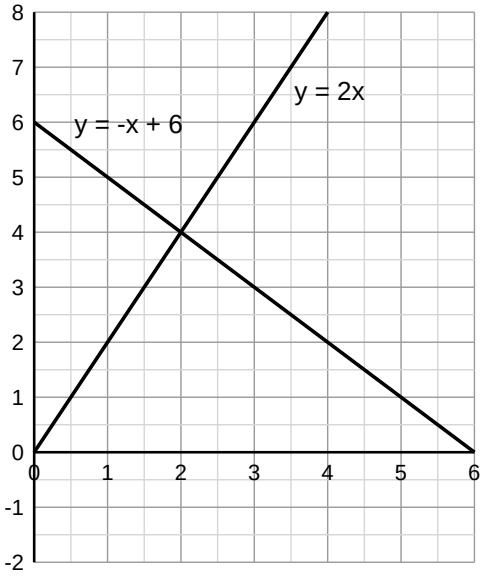
The graphs of $y = x - 1$ and $y = -x + 7$ are drawn on the grid.

Use the graphs to solve the simultaneous equations

$y = x - 1$

$y = -x + 7$

1.16



The graphs of $y = 2x$ and $y = -x + 6$ are drawn on the grid.

Use the graphs to solve the simultaneous equations

$y = 2x$

$y = -x + 6$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding with a number in front of x

Solve the simultaneous equations

$2x + y = 9$

$x - y = 3$

Step 1	$+y$ and $-y$: add	$3x = 12$
Step 2	Solve	$x = 4$
Step 3	Put x in	$4 - y = 3$
Step 4	Solve	$y = 1$

Practice: adding with a number in front of x

1.17 Solve the simultaneous equations

$2x + y = 12$

$x - y = 3$

1.18 Solve the simultaneous equations

$4x + y = 13$

$x - y = 2$

1.19 Solve the simultaneous equations

$3x + y = 11$

$x - y = -3$

1.20 Solve the simultaneous equations

$5x + y = 11$

$x - y = 1$

1.21 Solve the simultaneous equations

$2x + y = 15$

$x - y = 3$

1.22 Solve the simultaneous equations

$4x + y = 10$

$x - y = -5$